

MERCOR | James Moore | 13:00-13:20 ET | meet.google.com/gzy-xivr-vyu

James = research eng under Aksh, SkyRL training + environments. Peer. Ask what he works on, minute 1.

Clock: 0-2 pitch | 2-14 his Qs | 14-18 my Qs | 18-20 close. **≤90s per answer. Headline first. Stop.**

Leave him with: finds failures + builds the fix, with numbers | read their work incl. APEX 1.1 *yesterday* | gap = LLM RL plumbing, said plainly.

0 | LINES

- **Open:** "Hi James, good to meet you. Happy to start wherever you'd like."
- **Close:** "Thanks, this was useful. I'm keen. Next step and roughly when?"
- **Don't know:** "I don't know that. Here's what I'd check: ____." | **BMO stop:** "That's as far as I can go on internal detail."
- **Start:** "2-4 weeks from offer. Canadian citizen, TN at the border, no sponsorship."

1 | PITCH (90s)

1. **BMO:** bias to **downplay investment risk**, GenAI tool, **\$200B+ AUM** to deterministic eval pipeline, **hundreds of synthesized inputs**. Now: graph agent layer (**30,000 clients, 3h, was days**) + RL envs for wealth agents (starting).
2. **Merlyn:** **3 people, collective, nights/weekends. 8th BEHAVIOR-1K; proprioceptive collapse** (mask 60% to **+48%**). **LIBERO-PRO:** $\pi 0.5$ **96 to 21; recipe-induced, NOT architectural**; conservative FFT **21 to 42. RLinf** integration. Now: **Catan** agent env.
3. **Epineuron:** FDA Breakthrough neurostim, firmware/PCB. One sentence.
4. **Close verbatim:** "I treat systems like games: how they work, how they break, build the test, fix, repeat. Puzzles."

2 | LITMUS

5 skills | 120 tasks | groups of 8 | 5,760 rollouts. Learning $\propto k(8-k)/64$: max 4/8, **zero at 0 or 8**. Job: bet on ~50% tasks while all get easier.

Call	Why
Census epoch 1, one group each	blind; trains at uniform rate anyway
Newest reading, decay by age , never average	drift +0.12/epoch ; one group already at noise floor
Bonus for 1 & 2 of 8 ONLY, not 0	0/8 mostly never move (~38%). Draft hit 0s; Claude caught, I retargeted
Skill tilt (own credit tally) to biggest term	pass prob multiplies across skills; weak skill caps all. reasoning 13/120
Softmax not argmax	argmax hammers one 4/8 task
Skipped poison	honest pass, wrote threat model

Result: 16 held-out banks: **+0.029** (0.874 to 0.903), **16/16**, weakest skill **+0.058**. Flat-group waste **52% to 27%**. Stand-in test set rebuilt from gymsim band fn; salt = seed for a new bank.

Tools: Claude Code impl + sweeps. Mine: census, $p(1-p)$, bank hard tasks, staleness. Claude's: $k \in \{1,2\}$, softmax.

- $_w=0$: re-probe parked 0/8 when tally says skills moved. 16 banks, **monotonically worse**. Softmax already gives parked tasks a few % picks. **"Built it, measured it, lost, turned off, left the receipt."**
- **Poison +30min:** flipped sign, same size to hits my 4/8 farm; **picker = attack surface**; tally steers *away* from drained skill. Fix 1 **cap groups/task** (cost: retires a paying task). Fix 2 **quarantine** skill where tally says fed but tasks lag. $\text{Frac} \in [0,4]$, **can be 0 to must pay for itself**. Measure: salt sweep poison on/off.
- Budget: 960 +0 | 2880 **+0.058** | 5760 +.029 | 11520 +.010.
- **NEVER:** semaphore/backoff/parallel | DeepSeek | "ran over" (74 min) | bonus on 0.

3 | GAP "RL on an LLM?"

"Not on a language model." to RLinf integration; simple rejection sampling on $\pi 0.5$, **shallow** to BMO RL envs starting to Catan: rollouts to distill open weights to RL, outcome = verifier to "Litmus *is* this under a GRPO rule. Transfers: non-model errors to 0, no gain without ablation. Learn here: trainer/inference plumbing. Your logprob-mismatch check = haven't done."

Vocab: GRPO drops critic, $\text{adv}=(r-\text{mean})/\text{std}$ | DAPO: clip-higher, dynamic sampling, token loss, soft overlong | **prompt_mean +3.9** | DPPO masks/GLM-5 truncates, noise. **Stop where knowledge stops.**

4 | DEPTH

BMO | bias: advisors' tool; spot checks can't show "systematic"; synthesized inputs, deterministic to quantified skew | graph layer: harness w/ graph+analysis+search tools, multi-hop, 30k clients/3h | RL envs: virtual BMO work-worlds, experts write rubrics, R&D, "same shape as yours" | provisioning: agent from role+scope doc | to **stop line** | **never greeter robot**

Merlyn | BEHAVIOR: **22 of 50** trained, scored on 50, **10k demos**, OmniGibson; chunking beat ensembling **3x**; boundary resampling **doubled** long-tail | LIBERO-PRO: videos to arm to *old* location = memorization; FFT **b64, LR 1e-5** | **"lucky hparam?" to stable 8k-27k** | **matched std LIBERO** | **bounded** (b16/1e-6 to 26%) | LoRA **15-21 matched** | video prior **42 to 35** | paper: "wrote it, rejected CoRL, result stands" | RLinf: contributed integration, **didn't build it** | LessWrong = BEHAVIOR, **not** the paper

Catan | wanted env to Catanatron to bots **peeked hidden state** to rebuild native engine | view = full transcript + legal actions + negotiate | every action + reasoning saved | games played, **no numbers, compute-bound** | why: RL env + negotiation eval/alignment | **train:** save view/legal/tokens/action/msg to **1** rejection-SFT on winners to **2** branch saved states to pref pairs to **3** RL, outcome verifier (VP-delta gameable) | API = text not tokens to open weights | negotiation: judge promises kept/broken | TI only if asked

5 | WHY MERCOR + APEX 1.1 (bring up)

Their loop = mine, smaller. **397B**: harness alone **+6 pts** on 35B (22.7 to 28.7) before RL; 397B **16.1 to 27.3**; **file-diff grading hard to overfit** = my LIBERO-PRO lesson; "algorithm < data."

1.1, Sep 8: 80 tasks/domain, 3 audits | killed **scattergunning** (hedged multi-answers) | judge to DeepSeek-v4-Flash | info to files not sys prompt | **26,302 trajectories** scanned | **Pass^4. Say**: "same bug class as file-diff: grade a committed artifact, not narration."

6 | ASK (pick 2)

1. "Harness alone moved 35B ~6 pts. Team time now: harness/verifier vs training loop?"
2. "1.1 yesterday: caught scattergunning via trajectory review or Pass^4 dropping? 'Grade the artifact' a design rule now?"
3. "How does Post-Training split from Benchmarking / Real Environments siblings? Normal week?"

7 | DON'T

"published"/"under review" | "startup" | robot | "US-authorized" | "sort out visa" | VLM judges | "50/60/100%" (**48%**) | "learning decay" | comp (if asked: "your reqs show \$180-500K+equity, where's this?") | Charlie | fill silence | "adaptable/always learning" | "we" for solo